

# Equity Preservation Intensity and the Cross Section of Stock Returns

I. M. Harking

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## Abstract

This paper studies the asset pricing implications of Equity Preservation Intensity (EPI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on EPI achieves an annualized gross (net) Sharpe ratio of 0.51 (0.44), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 19 (19) bps/month with a t-statistic of 2.43 (2.54), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, Net equity financing) is 19 bps/month with a t-statistic of 2.43.

# 1 Introduction

The cross-section of expected returns continues to challenge our understanding of how investors price consumption risk in financial markets. While classical asset pricing theory predicts that securities with higher exposure to systematic consumption risk should command higher expected returns, empirical evidence reveals persistent patterns that standard models struggle to explain. We identify a novel predictor of stock returns based on firms' equity preservation intensity (EPI), defined as the negative change in common stock relative to the change in total assets. This measure captures firms' propensity to preserve equity capital during periods of economic uncertainty, potentially signaling their exposure to aggregate consumption risk and investors' time-varying marginal utility of wealth.

From a consumption-based asset pricing perspective, firms with high equity preservation intensity exhibit characteristics that make them particularly sensitive to consumption risk. When firms actively preserve equity capital by limiting dilution and maintaining stable equity bases, they signal lower flexibility in responding to adverse consumption shocks that affect investors' marginal utility [Campbell and Cochrane \(1999\)](#). These firms face binding constraints during bad states of the world when the marginal utility of consumption is high, making their cash flows more procyclical with consumption growth [Lettau and Ludvigson \(2001\)](#). The preservation of equity capital thus exposes investors to heightened sensitivity to long-run consumption risks, as these firms cannot easily adjust their capital structure to smooth shocks that correlate with aggregate consumption [Bansal and Yaron \(2004\)](#).

The state-dependent nature of equity preservation intensity links directly to the intertemporal marginal rate of substitution. During economic downturns characterized by low consumption growth, firms that maintain high equity preservation intensity face amplified operating leverage effects that increase their covariance with the stochastic discount factor [Zhang \(2005\)](#). This mechanism operates through the inter-

action between financial constraints and consumption disasters—when consumption growth is low or negative, equity-preserving firms experience disproportionate value destruction because they cannot optimally adjust their financing mix [Barro \(2006\)](#). The inability to issue equity during precisely those states when external financing is most valuable creates an additional risk premium demanded by investors with habit-formation preferences [Campbell and Cochrane \(1999\)](#).

Furthermore, equity preservation intensity captures firms’ exposure to systematic variation in the consumption-wealth ratio. High EPI firms exhibit limited financial flexibility that prevents them from exploiting time-varying investment opportunities when the price of consumption risk changes [Merton \(1973\)](#). This inflexibility becomes particularly costly during periods of high marginal utility, when investors most value the option to reallocate capital across states [Cochrane and Piazzesi \(2005\)](#). The consumption-based pricing kernel thus assigns higher loadings to firms with elevated equity preservation intensity, as their returns covary more strongly with innovations to long-run expected consumption growth and cannot be easily hedged through dynamic portfolio strategies [Parker and Julliard \(2005\)](#).

We find strong evidence that equity preservation intensity predicts the cross-section of stock returns. The value-weighted long/short EPI strategy achieves an annualized gross Sharpe ratio of 0.51 and generates a monthly average excess return of 32 basis points with a t-statistic of 3.96. The strategy’s alpha relative to the Fama-French six-factor model equals 19 basis points per month with a t-statistic of 2.43, demonstrating that EPI captures a distinct dimension of systematic risk not explained by existing factors. The economic magnitude of these returns confirms that investors demand substantial compensation for bearing the consumption risk embedded in high EPI stocks.

The robustness of our findings extends across alternative portfolio construction methodologies and persists after accounting for transaction costs. Net of trading

costs, the EPI strategy maintains an annualized Sharpe ratio of 0.44 and delivers a monthly alpha of 19 basis points relative to the six-factor model with a t-statistic of 2.54. Among the largest quintile of stocks by market capitalization, where implementation is most feasible, the strategy generates average returns of 23 basis points per month with a t-statistic of 2.39. These results indicate that the consumption risk premium associated with equity preservation intensity is not confined to small, illiquid stocks but represents a pervasive pricing phenomenon across the market capitalization spectrum.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the EPI strategy maintains an alpha of 19 basis points per month with a t-statistic of 2.43. The strategy’s gross Sharpe ratio of 0.51 exceeds 90% of documented anomalies in the factor zoo, while its net Sharpe ratio of 0.44 surpasses 97% of existing strategies. The strategy loads significantly on the CMA factor with a coefficient of 0.40 and t-statistic of 7.39, consistent with our hypothesis that equity preservation intensity captures firms’ differential exposure to investment frictions that amplify consumption risk during bad states.

Our paper contributes to the consumption-based asset pricing literature by identifying a novel cross-sectional predictor that directly links firms’ financing decisions to their exposure to aggregate consumption risk. While [Fama and French \(2015\)](#) document that investment and profitability factors help explain average returns, and [Hou et al. \(2015\)](#) propose a q-factor model based on investment theory, our equity preservation intensity measure provides a consumption-based rationale for why certain financing patterns generate systematic risk premia. Unlike mechanical accounting variables studied in [Cooper et al. \(2008\)](#) and [Daniel and Titman \(2006\)](#), EPI captures the economic channel through which financial inflexibility amplifies firms’ covariance with the stochastic discount factor during high marginal utility states.

Methodologically, we advance the literature by demonstrating that equity preser-

vation intensity improves the mean-variance efficient frontier even after controlling for comprehensive sets of existing factors and anomalies. Following the rigorous testing protocol of [Novy-Marx and Velikov \(2023b\)](#), we show that EPI’s predictive power is not subsumed by related measures of equity issuance or payout policy studied in [Pontiff and Woodgate \(2008\)](#) and [Ikenberry et al. \(1995\)](#). The alpha of 19 basis points per month relative to twelve factors—the six Fama-French factors plus six closely related anomalies—with a t-statistic of 2.43 establishes that EPI captures a distinct dimension of consumption risk. Our net Sharpe ratio analysis reveals that EPI ranks in the top 3% of strategies after accounting for implementation costs, addressing concerns raised by [Novy-Marx and Velikov \(2016a\)](#) about the tradability of documented anomalies.

Our findings have broader implications for understanding how financial frictions interact with consumption-based pricing kernels to generate cross-sectional return variation. The success of equity preservation intensity in predicting returns supports models where firms’ financing constraints bind most severely during consumption disasters, creating state-dependent risk exposures that rational investors price [Rietz \(1988\)](#). By documenting that high EPI firms command return premia consistent with elevated consumption risk exposure, we provide empirical support for theoretical frameworks linking corporate financial policy to systematic risk through the lens of investors’ marginal utility. These results suggest that accounting-based measures of financial flexibility contain valuable information about firms’ fundamental risk characteristics, offering new avenues for testing and refining consumption-based asset pricing models.

## 2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in COMPUSTAT, spanning several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of equity issuance activity and depreciation expenses across firms. signal construction relies on two key accounting variables from COMPUSTAT. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the balance sheet, reflecting the book value of common equity at par. Depreciation, depletion, and amortization (DPACT) captures the total depreciation, depletion, and amortization expense reported on the cash flow statement, representing the systematic allocation of tangible and intangible asset costs over their useful lives. These variables provide essential information about firms' equity capital structure and their ongoing capital consumption patterns. construct the Equity Preservation Intensity signal by calculating the year-over-year change in common stock scaled by lagged depreciation expense:  $(CSTK(t) - CSTK(t-1)) / DPACT(t-1)$ . This ratio captures the extent to which firms issue new common equity relative to their rate of capital consumption, as proxied by depreciation expenses. A higher value indicates more intensive equity issuance activity relative to the firm's asset depreciation rate, potentially signaling management's efforts to preserve or strengthen the equity base. We compute this measure using fiscal year-end values and require non-missing values for both CSTK and DPACT to ensure data quality. To mitigate the influence of extreme outliers, we winsorize the signal at the 1st and 99th percentiles within each year.

### 3 Signal diagnostics

Figure 1 plots descriptive statistics for the EPI signal. Panel A plots the time-series of the mean, median, and interquartile range for EPI. On average, the cross-sectional mean (median) EPI is -0.21 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input EPI data. The signal's interquartile range spans -0.03 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the EPI signal for the CRSP universe. On average, the EPI signal is available for 5.72% of CRSP names, which on average make up 7.11% of total market capitalization.

### 4 Does EPI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EPI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EPI portfolio and sells the low EPI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short EPI strategy earns an average return of 0.32% per month with a t-statistic of 3.96. The annualized Sharpe ratio of the strategy is 0.51. The alphas range from 0.19% to 0.37% per month and have t-statistics exceeding 2.43 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.40,

with a t-statistic of 7.39 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 485 stocks and an average market capitalization of at least \$1,381 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 23 bps/month with a t-statistics of 3.00. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for thirteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016b\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns



reported in the first column range between 20-35bps/month. The lowest return, (20 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.54. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EPI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-four cases.

Table 3 provides direct tests for the role size plays in the EPI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EPI, as well as average returns and alphas for long/short trading EPI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80<sup>th</sup> NYSE percentile), the EPI strategy achieves an average return of 23 bps/month with a t-statistic of 2.39. Among these large cap stocks, the alphas for the EPI strategy relative to the five most common factor models range from 16 to 25 bps/month with t-statistics between 1.61 and 2.64.

## 5 How does EPI perform relative to the zoo?

Figure 2 puts the performance of EPI in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.<sup>1</sup> The vertical red line shows where the Sharpe ratio for the EPI strategy falls in the distribution. The EPI strategy’s gross (net) Sharpe ratio of 0.51 (0.44) is greater than 90% (97%) of anomaly Sharpe ratios,

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<sup>1</sup>The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EPI strategy (red line).<sup>2</sup> Ignoring trading costs, a \$1 invested in the EPI strategy would have yielded \$8.09 which ranks the EPI strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EPI strategy would have yielded \$5.86 which ranks the EPI strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016b) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EPI relative to those. Panel A shows that the EPI strategy gross alphas fall between the 64 and 71 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EPI strategy has a positive net generalized alpha for five out of the five factor models. In these cases EPI ranks between the 83 and 89 percentiles in terms of how much it could have expanded the achievable investment frontier.

## 6 Does EPI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently

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<sup>2</sup>The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EPI with 210 filtered anomaly signals.<sup>3</sup> Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EPI or at least to weaken the power EPI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EPI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on  $\beta_{EPI}$  from Fama-MacBeth regressions of the form  $r_{i,t} = \alpha + \beta_{EPI}EPI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$ , where  $X$  stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on  $\alpha$  from spanning tests of the form:  $r_{EPI,t} = \alpha + \beta r_{X,t} + \epsilon_t$ , where  $r_{X,t}$  stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EPI. Stocks are finally grouped into five EPI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EPI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EPI and the six anomalies most closely-related to it. The six most-closely related anomalies

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<sup>3</sup>When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EPI signal in these Fama-MacBeth regressions exceed 2.07, with the minimum t-statistic occurring when controlling for Net Payout Yield. Controlling for all six closely related anomalies, the t-statistic on EPI is 1.90.

Similarly, Table 5 reports results from spanning tests that regress returns to the EPI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EPI strategy earns alphas that range from 12-23bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.63, which is achieved when controlling for Net Payout Yield. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EPI trading strategy achieves an alpha of 19bps/month with a t-statistic of 2.43.

## 7 Does EPI add relative to the whole zoo?

Finally, we can ask how much adding EPI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the EPI signal.<sup>4</sup> We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors,

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<sup>4</sup>We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EPI is available.

and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes EPI grows to \$2840.66.

## 8 Conclusion

This study provides robust evidence that Equity Preservation Intensity (EPI) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol of Novy-Marx and Velikov (2023), demonstrates that EPI-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies.

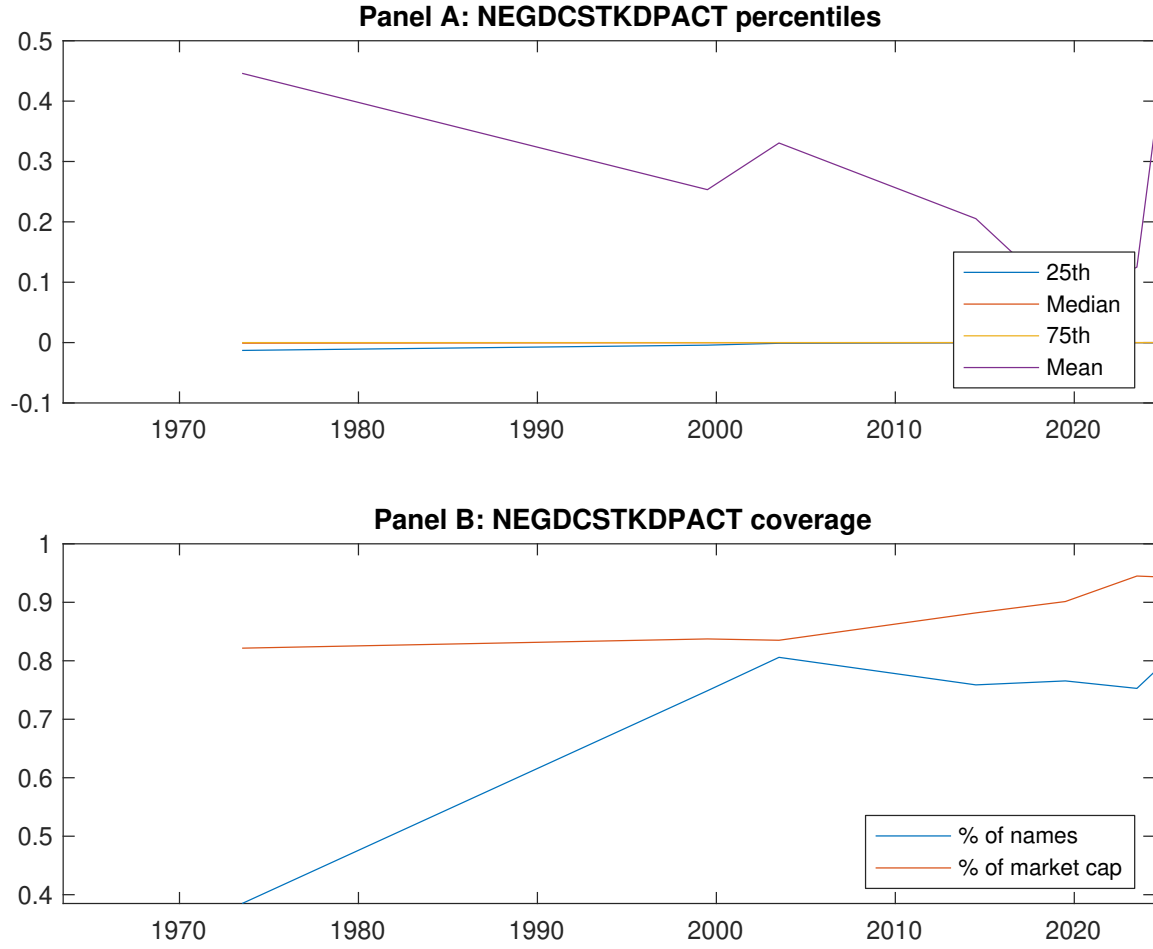
The key findings of our research underscore the economic importance of EPI as an asset pricing signal. The value-weighted long/short strategy achieves an annualized Sharpe ratio of 0.51 gross and 0.44 net of transaction costs, indicating that the signal remains profitable even after accounting for implementation frictions. Most notably, the strategy delivers a statistically significant monthly alpha of 19 basis points ( $t$ -statistic = 2.43) relative to the Fama-French five-factor model augmented with momentum, demonstrating that EPI captures a distinct source of return variation not explained by conventional risk factors. The persistence of this alpha even after controlling for six closely related equity financing strategies—including share issuance measures, net payout yield, and equity financing metrics—further validates EPI as an independent pricing signal rather than a mere repackaging of known anomalies.

From a practical perspective, these results suggest that investors can potentially enhance portfolio performance by incorporating EPI into their investment processes. The signal’s robustness to transaction costs and its orthogonality to existing factors

make it particularly attractive for quantitative investment strategies seeking diversified sources of alpha. Moreover, the economic magnitude of the returns suggests that market inefficiencies related to firms' equity preservation behaviors remain exploitable, even in increasingly efficient markets.

However, several limitations warrant consideration. First, while our analysis controls for major risk factors and related strategies, the possibility remains that EPI proxies for an unidentified systematic risk factor that justifies the return premium. Second, the implementation of EPI strategies in practice may face capacity constraints and market impact costs beyond those captured in our transaction cost estimates. Third, our sample period and geographic focus may limit the generalizability of findings to other markets and time periods.

Future research should explore several promising avenues. Investigating the economic mechanisms underlying the EPI effect could provide deeper insights into why markets appear to misprice equity preservation behaviors. International evidence would help establish whether the EPI premium represents a global phenomenon or is specific to certain market structures. Additionally, examining the time-varying nature of the EPI premium and its relationship to macroeconomic conditions could enhance our understanding of when the strategy performs best. Finally, exploring potential interactions between EPI and corporate governance characteristics or institutional ownership might reveal important cross-sectional variations in the signal's effectiveness. Such investigations would not only advance our theoretical understanding of asset pricing but also refine the practical implementation of EPI-based investment strategies.



**Figure 1:** Times series of EPI percentiles and coverage. This figure plots descriptive statistics for EPI. Panel A shows cross-sectional percentiles of EPI over the sample. Panel B plots the monthly coverage of EPI relative to the universe of CRSP stocks with available market capitalizations.

**Table 1:** Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EPI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, [Fama and French \(1993\)](#) three-factor model, [Fama and French \(1993\)](#) three-factor model augmented with the [Carhart \(1997\)](#) momentum factor, [Fama and French \(2015\)](#) five-factor model, and the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor following [Fama and French \(2018\)](#). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the [Fama and French \(2015\)](#) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

| Panel A: Excess returns and alphas on EPI-sorted portfolios                                       |                  |                  |                  |                  |                  |                  |
|---------------------------------------------------------------------------------------------------|------------------|------------------|------------------|------------------|------------------|------------------|
|                                                                                                   | (L)              | (2)              | (3)              | (4)              | (H)              | (H-L)            |
| $r^e$                                                                                             | 0.44<br>[2.55]   | 0.57<br>[3.22]   | 0.66<br>[3.80]   | 0.63<br>[3.92]   | 0.76<br>[4.80]   | 0.32<br>[3.96]   |
| $\alpha_{CAPM}$                                                                                   | -0.15<br>[-2.96] | -0.04<br>[-0.78] | 0.07<br>[1.36]   | 0.08<br>[1.73]   | 0.22<br>[4.54]   | 0.37<br>[4.66]   |
| $\alpha_{FF3}$                                                                                    | -0.11<br>[-2.28] | -0.00<br>[-0.05] | 0.09<br>[1.84]   | 0.04<br>[0.99]   | 0.21<br>[4.28]   | 0.32<br>[4.10]   |
| $\alpha_{FF4}$                                                                                    | -0.09<br>[-1.71] | 0.02<br>[0.32]   | 0.07<br>[1.50]   | 0.04<br>[0.82]   | 0.19<br>[3.80]   | 0.28<br>[3.45]   |
| $\alpha_{FF5}$                                                                                    | -0.10<br>[-1.97] | 0.03<br>[0.69]   | 0.07<br>[1.35]   | -0.05<br>[-1.19] | 0.12<br>[2.53]   | 0.22<br>[2.83]   |
| $\alpha_{FF6}$                                                                                    | -0.08<br>[-1.55] | 0.05<br>[0.95]   | 0.06<br>[1.11]   | -0.05<br>[-1.08] | 0.11<br>[2.33]   | 0.19<br>[2.43]   |
| Panel B: <a href="#">Fama and French (2018)</a> 6-factor model loadings for EPI-sorted portfolios |                  |                  |                  |                  |                  |                  |
| $\beta_{MKT}$                                                                                     | 0.98<br>[80.20]  | 1.00<br>[88.10]  | 1.01<br>[83.13]  | 1.00<br>[96.11]  | 0.97<br>[82.65]  | -0.01<br>[-0.52] |
| $\beta_{SMB}$                                                                                     | 0.01<br>[0.37]   | 0.04<br>[2.50]   | 0.04<br>[2.36]   | -0.07<br>[-4.82] | -0.02<br>[-1.12] | -0.03<br>[-0.93] |
| $\beta_{HML}$                                                                                     | -0.04<br>[-1.87] | -0.11<br>[-4.89] | -0.08<br>[-3.47] | -0.00<br>[-0.17] | -0.06<br>[-2.66] | -0.02<br>[-0.44] |
| $\beta_{RMW}$                                                                                     | 0.05<br>[2.06]   | -0.08<br>[-3.66] | 0.05<br>[2.00]   | 0.11<br>[5.36]   | 0.12<br>[5.09]   | 0.07<br>[1.82]   |
| $\beta_{CMA}$                                                                                     | -0.15<br>[-4.31] | -0.02<br>[-0.58] | 0.03<br>[0.79]   | 0.30<br>[10.21]  | 0.25<br>[7.48]   | 0.40<br>[7.39]   |
| $\beta_{UMD}$                                                                                     | -0.03<br>[-2.38] | -0.02<br>[-1.62] | 0.02<br>[1.35]   | -0.01<br>[-0.56] | 0.01<br>[1.01]   | 0.04<br>[2.15]   |
| Panel C: Average number of firms ( $n$ ) and market capitalization ( $me$ )                       |                  |                  |                  |                  |                  |                  |
| $n$                                                                                               | 731              | 605              | 485              | 592              | 659              |                  |
| $me$ (\$10 <sup>6</sup> )                                                                         | 1630             | 1381             | 1825             | 2022             | 2427             |                  |



**Table 2:** Robustness to sorting methodology & trading costs

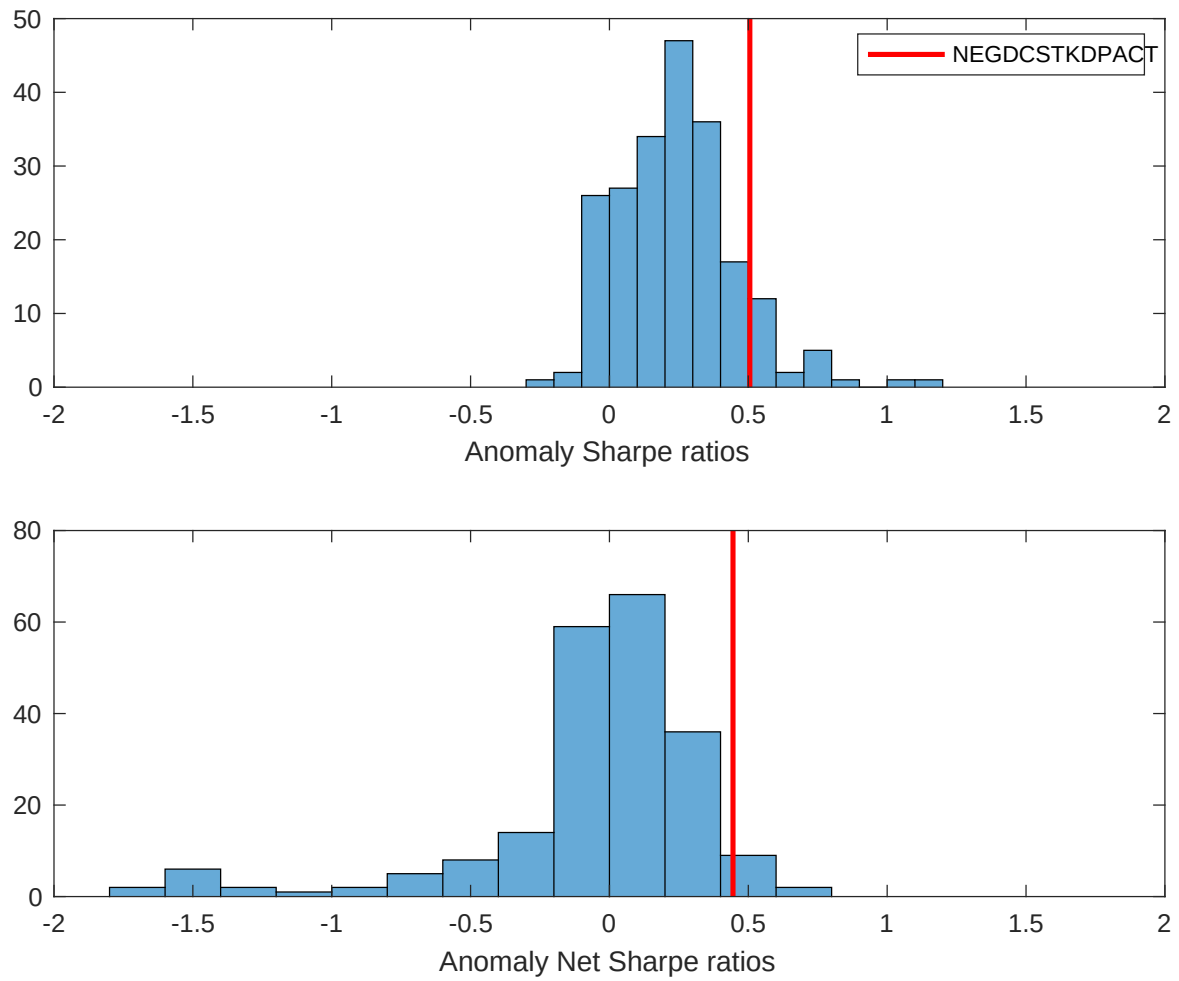
This table evaluates the robustness of the choices made in the EPI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016b) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

| Panel A: Gross Returns and Alphas                                         |        |         |                    |                          |                         |                         |                         |                         |
|---------------------------------------------------------------------------|--------|---------|--------------------|--------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
| Portfolios                                                                | Breaks | Weights | $r^e$              | $\alpha_{\text{CAPM}}$   | $\alpha_{\text{FF3}}$   | $\alpha_{\text{FF4}}$   | $\alpha_{\text{FF5}}$   | $\alpha_{\text{FF6}}$   |
| Quintile                                                                  | NYSE   | VW      | 0.32<br>[3.96]     | 0.37<br>[4.66]           | 0.32<br>[4.10]          | 0.28<br>[3.45]          | 0.22<br>[2.83]          | 0.19<br>[2.43]          |
| Quintile                                                                  | NYSE   | EW      | 0.55<br>[6.96]     | 0.65<br>[8.83]           | 0.54<br>[8.36]          | 0.44<br>[6.96]          | 0.36<br>[6.08]          | 0.29<br>[5.00]          |
| Quintile                                                                  | Name   | VW      | 0.29<br>[3.61]     | 0.34<br>[4.14]           | 0.28<br>[3.50]          | 0.24<br>[2.93]          | 0.19<br>[2.39]          | 0.16<br>[2.05]          |
| Quintile                                                                  | Cap    | VW      | 0.23<br>[3.00]     | 0.28<br>[3.59]           | 0.24<br>[3.12]          | 0.20<br>[2.50]          | 0.17<br>[2.20]          | 0.14<br>[1.80]          |
| Decile                                                                    | NYSE   | VW      | 0.32<br>[3.32]     | 0.36<br>[3.80]           | 0.28<br>[2.97]          | 0.23<br>[2.46]          | 0.23<br>[2.49]          | 0.20<br>[2.15]          |
| Panel B: Net Returns and Novy-Marx and Velikov (2016b) generalized alphas |        |         |                    |                          |                         |                         |                         |                         |
| Portfolios                                                                | Breaks | Weights | $r_{\text{net}}^e$ | $\alpha_{\text{CAPM}}^*$ | $\alpha_{\text{FF3}}^*$ | $\alpha_{\text{FF4}}^*$ | $\alpha_{\text{FF5}}^*$ | $\alpha_{\text{FF6}}^*$ |
| Quintile                                                                  | NYSE   | VW      | 0.28<br>[3.49]     | 0.34<br>[4.24]           | 0.29<br>[3.68]          | 0.26<br>[3.32]          | 0.21<br>[2.75]          | 0.19<br>[2.54]          |
| Quintile                                                                  | NYSE   | EW      | 0.35<br>[4.10]     | 0.47<br>[5.85]           | 0.36<br>[5.14]          | 0.31<br>[4.49]          | 0.20<br>[3.02]          | 0.16<br>[2.60]          |
| Quintile                                                                  | Name   | VW      | 0.25<br>[3.13]     | 0.30<br>[3.67]           | 0.24<br>[3.05]          | 0.22<br>[2.74]          | 0.17<br>[2.24]          | 0.16<br>[2.06]          |
| Quintile                                                                  | Cap    | VW      | 0.20<br>[2.54]     | 0.24<br>[3.13]           | 0.20<br>[2.65]          | 0.18<br>[2.31]          | 0.15<br>[2.03]          | 0.14<br>[1.80]          |
| Decile                                                                    | NYSE   | VW      | 0.27<br>[2.86]     | 0.31<br>[3.31]           | 0.23<br>[2.53]          | 0.21<br>[2.26]          | 0.20<br>[2.22]          | 0.19<br>[2.05]          |

**Table 3:** Conditional sort on size and EPI

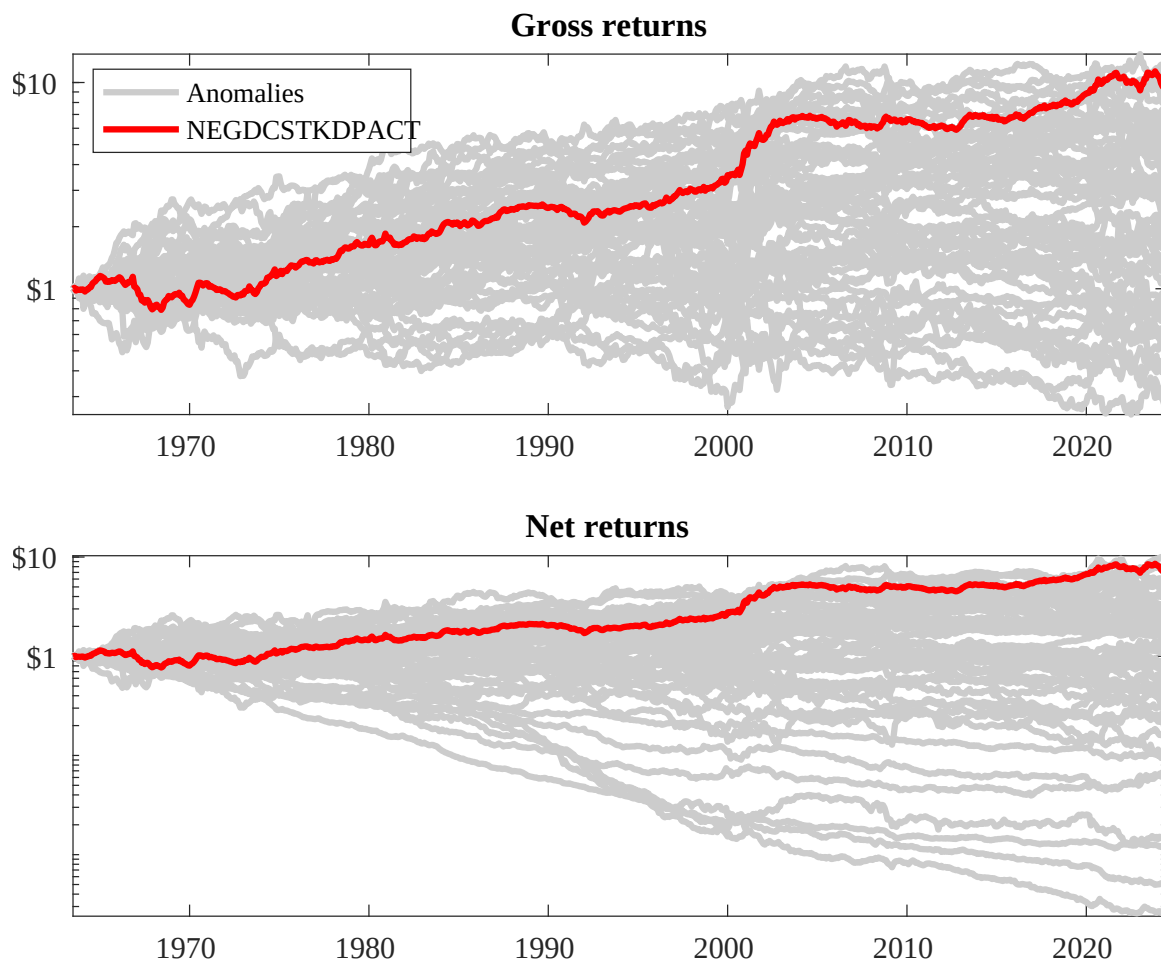
This table presents results for conditional double sorts on size and EPI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EPI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EPI and short stocks with low EPI. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

| Panel A: portfolio average returns and time-series regression results |               |        |        |        |        |                                                    |        |                 |                |                |                |                |
|-----------------------------------------------------------------------|---------------|--------|--------|--------|--------|----------------------------------------------------|--------|-----------------|----------------|----------------|----------------|----------------|
| Size quintiles                                                        | EPI Quintiles |        |        |        |        | EPI Strategies                                     |        |                 |                |                |                |                |
|                                                                       |               | (L)    | (2)    | (3)    | (4)    | (H)                                                | $r^e$  | $\alpha_{CAPM}$ | $\alpha_{FF3}$ | $\alpha_{FF4}$ | $\alpha_{FF5}$ | $\alpha_{FF6}$ |
|                                                                       | (1)           | 0.39   | 0.72   | 0.95   | 0.96   | 0.97                                               | 0.60   | 0.69            | 0.59           | 0.52           | 0.38           | 0.33           |
|                                                                       |               | [1.39] | [2.73] | [3.61] | [3.89] | [4.11]                                             | [5.44] | [6.47]          | [6.06]         | [5.30]         | [4.17]         | [3.64]         |
|                                                                       | (2)           | 0.51   | 0.77   | 0.87   | 0.96   | 0.92                                               | 0.41   | 0.51            | 0.38           | 0.32           | 0.27           | 0.23           |
|                                                                       |               | [2.04] | [3.22] | [3.65] | [4.17] | [4.21]                                             | [3.97] | [5.14]          | [4.21]         | [3.54]         | [3.00]         | [2.55]         |
|                                                                       | (3)           | 0.62   | 0.58   | 0.80   | 0.83   | 0.94                                               | 0.32   | 0.39            | 0.29           | 0.25           | 0.19           | 0.17           |
|                                                                       | [2.86]        | [2.72] | [3.58] | [4.09] | [4.79] | [3.54]                                             | [4.46] | [3.52]          | [3.05]         | [2.30]         | [2.04]         |                |
| (4)                                                                   | 0.50          | 0.61   | 0.85   | 0.75   | 0.79   | 0.28                                               | 0.35   | 0.23            | 0.21           | 0.04           | 0.04           |                |
|                                                                       | [2.55]        | [3.03] | [4.28] | [3.96] | [4.35] | [3.27]                                             | [4.02] | [2.96]          | [2.68]         | [0.52]         | [0.55]         |                |
| (5)                                                                   | 0.49          | 0.49   | 0.53   | 0.51   | 0.71   | 0.23                                               | 0.25   | 0.22            | 0.18           | 0.18           | 0.16           |                |
|                                                                       | [2.94]        | [2.78] | [3.10] | [3.14] | [4.53] | [2.39]                                             | [2.64] | [2.32]          | [1.89]         | [1.90]         | [1.61]         |                |
| Panel B: Portfolio average number of firms and market capitalization  |               |        |        |        |        |                                                    |        |                 |                |                |                |                |
| Size quintiles                                                        | EPI Quintiles |        |        |        |        | EPI Quintiles                                      |        |                 |                |                |                |                |
|                                                                       | Average $n$   |        |        |        |        | Average market capitalization (\$10 <sup>6</sup> ) |        |                 |                |                |                |                |
|                                                                       |               | (L)    | (2)    | (3)    | (4)    | (H)                                                | (L)    | (2)             | (3)            | (4)            | (H)            |                |
|                                                                       | (1)           | 341    | 342    | 342    | 340    | 334                                                | 26     | 29              | 33             | 25             | 24             |                |
|                                                                       | (2)           | 97     | 95     | 96     | 96     | 94                                                 | 50     | 49              | 51             | 50             | 48             |                |
|                                                                       | (3)           | 69     | 69     | 69     | 69     | 69                                                 | 86     | 85              | 87             | 88             | 89             |                |
|                                                                       | (4)           | 60     | 60     | 60     | 60     | 60                                                 | 186    | 189             | 199            | 196            | 200            |                |
| (5)                                                                   | 56            | 56     | 56     | 56     | 55     | 1308                                               | 1408   | 1476            | 1529           | 1779           |                |                |



**Figure 2:** Distribution of Sharpe ratios.

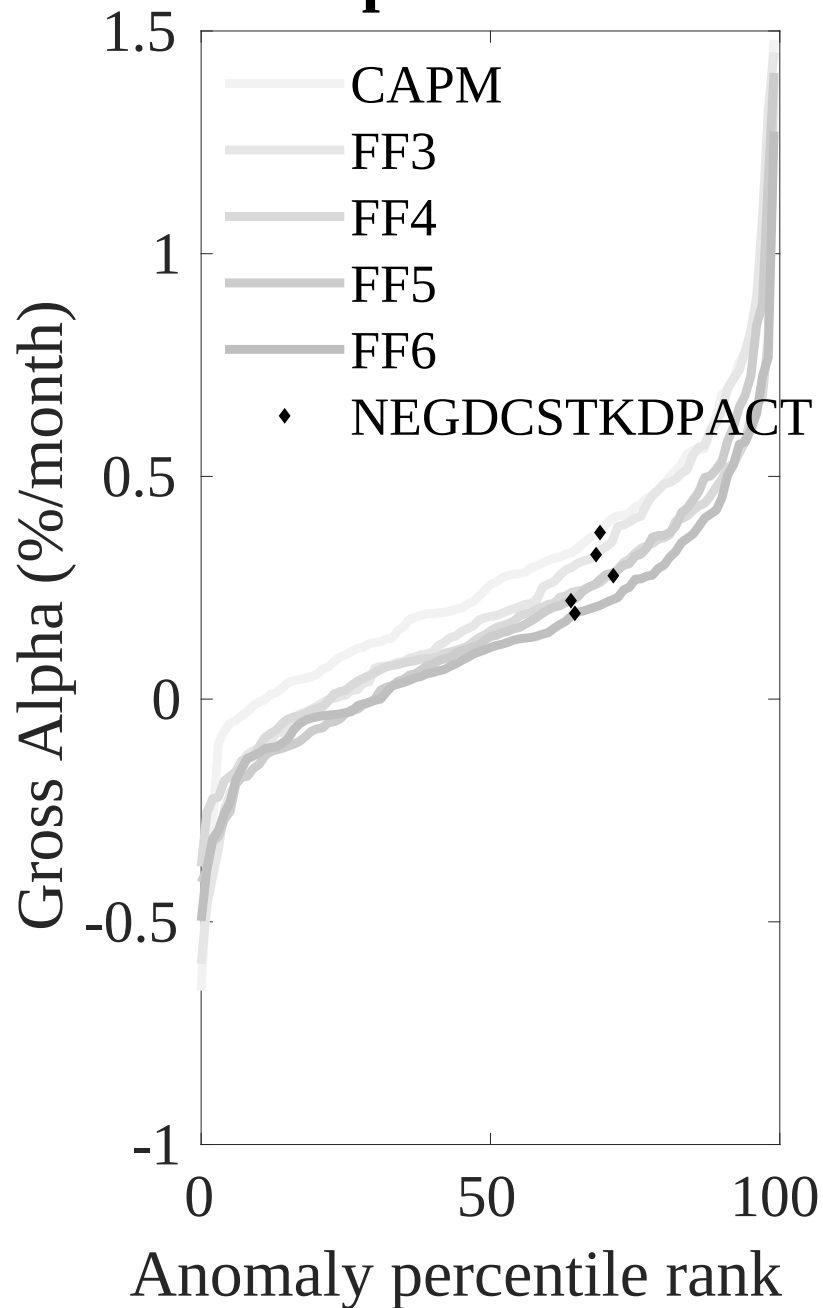
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EPI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.



**Figure 3:** Dollar invested.

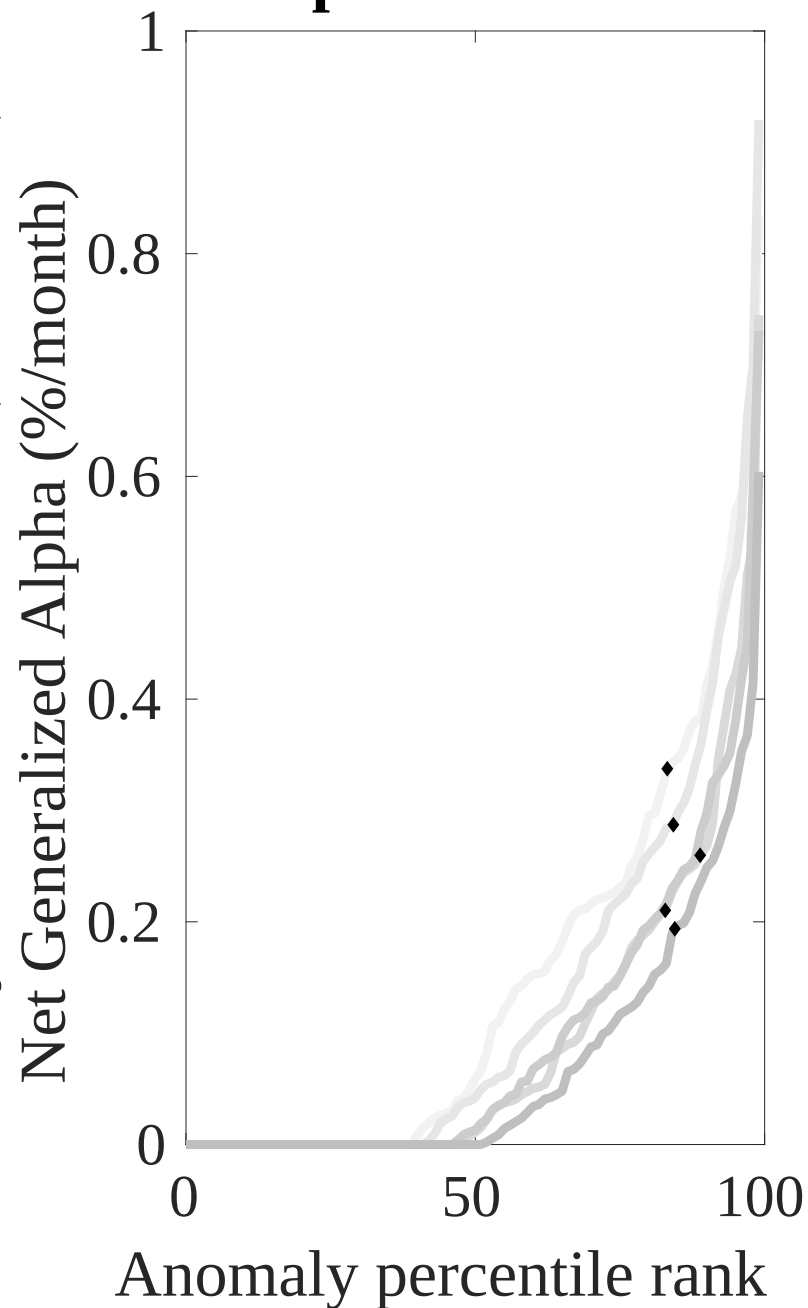
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EPI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

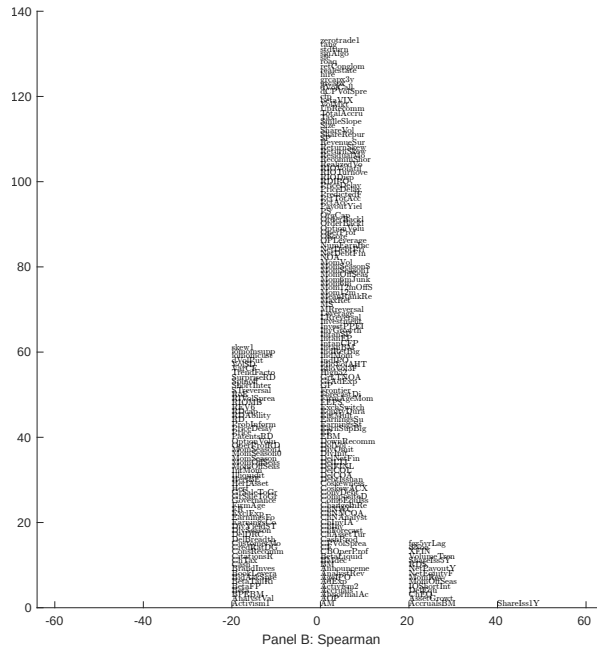
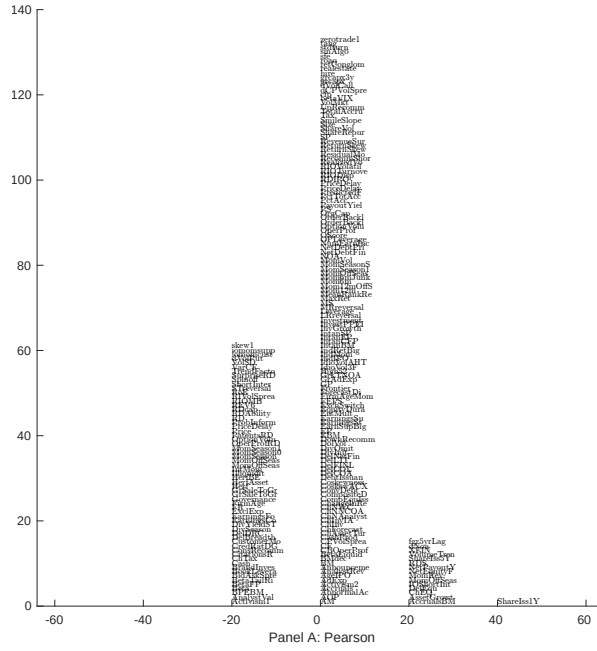
## Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

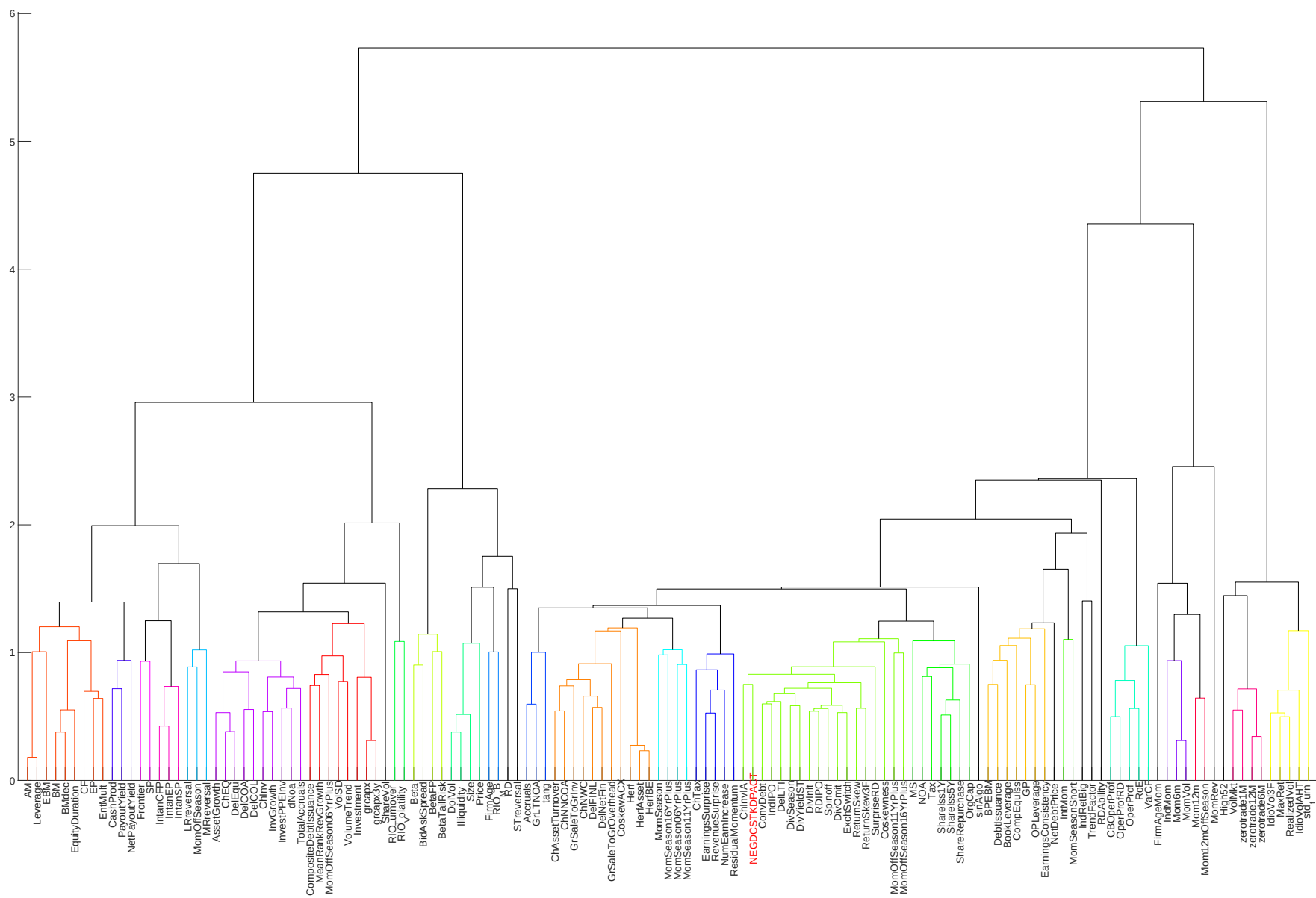
## Net Alpha Percentiles





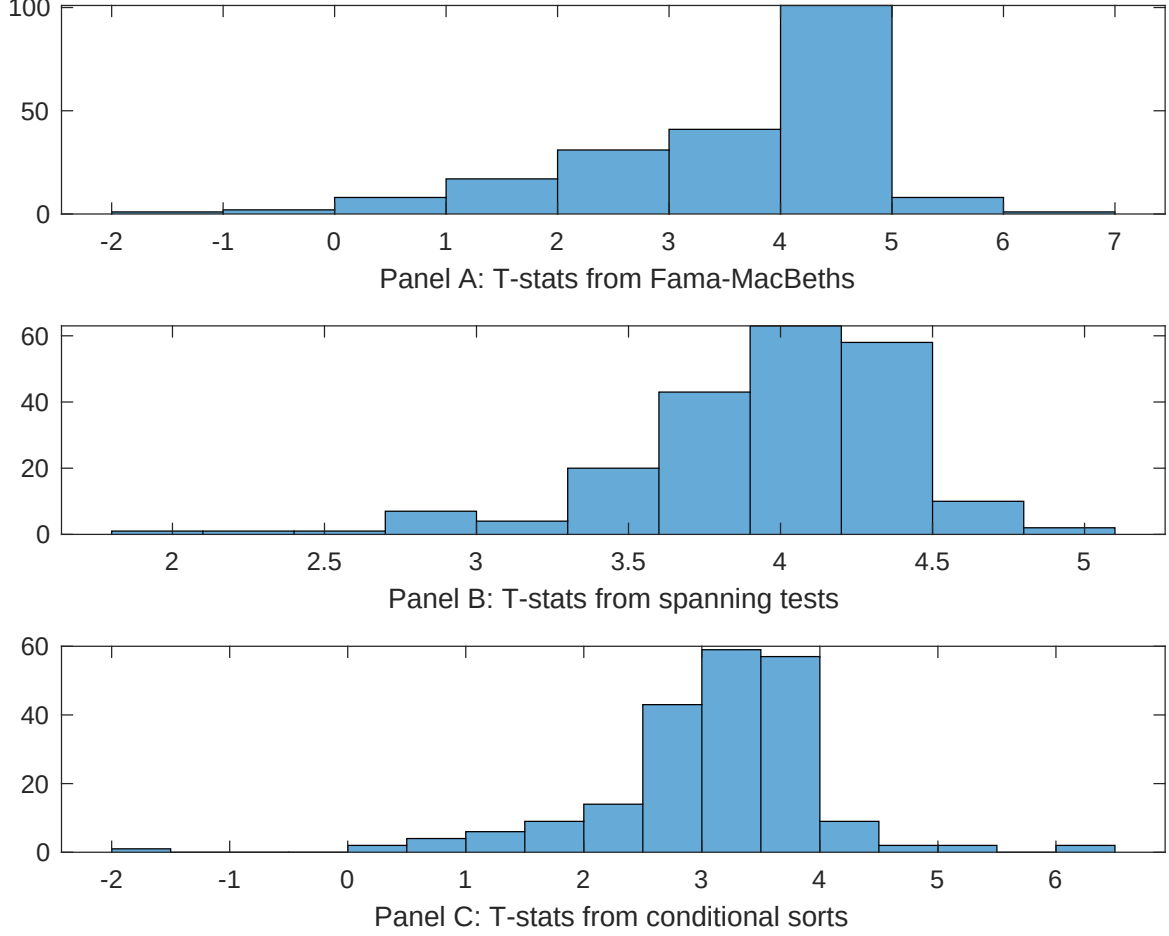
**Figure 5:** Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with EPI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.



**Figure 6:** Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.



**Figure 7:** Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of EPI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on  $\beta_{EPI}$  from Fama-MacBeth regressions of the form  $r_{i,t} = \alpha + \beta_{EPI}EPI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$ , where  $X$  stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on  $\alpha$  from spanning tests of the form:  $r_{EPI,t} = \alpha + \beta r_{X,t} + \epsilon_t$ , where  $r_{X,t}$  stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EPI. Stocks are finally grouped into five EPI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EPI trading strategies conditioned on each of the 210 filtered anomalies.



**Table 4:** Fama-MacBeths controlling for most closely related anomalies

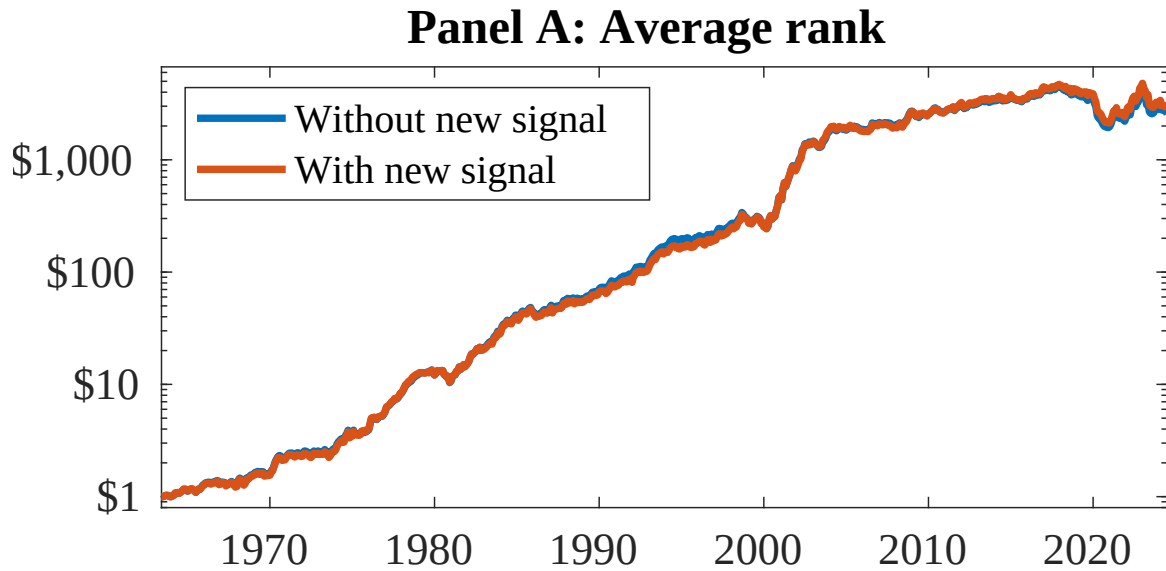
This table presents Fama-MacBeth results of returns on EPI. and the six most closely related anomalies. The regressions take the following form:  $r_{i,t} = \alpha + \beta_{EPI}EPI_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$ . The six most closely related anomalies,  $X$ , are Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

|                 |                |                |                |                |                |                |                  |
|-----------------|----------------|----------------|----------------|----------------|----------------|----------------|------------------|
| Intercept       | 0.13<br>[5.98] | 0.12<br>[5.43] | 0.17<br>[6.64] | 0.13<br>[6.37] | 0.13<br>[5.61] | 0.13<br>[5.42] | 0.14<br>[5.00]   |
| EPI             | 0.74<br>[3.87] | 0.45<br>[2.07] | 0.71<br>[3.32] | 0.69<br>[3.66] | 0.81<br>[3.92] | 0.75<br>[3.47] | 0.44<br>[1.90]   |
| Anomaly 1       | 0.14<br>[6.01] |                |                |                |                |                | 0.85<br>[3.78]   |
| Anomaly 2       |                | 0.21<br>[1.88] |                |                |                |                | 0.12<br>[2.60]   |
| Anomaly 3       |                |                | 0.38<br>[3.21] |                |                |                | -0.25<br>[-0.16] |
| Anomaly 4       |                |                |                | 0.27<br>[4.73] |                |                | 0.27<br>[5.88]   |
| Anomaly 5       |                |                |                |                | 0.12<br>[3.17] |                | 0.98<br>[1.63]   |
| Anomaly 6       |                |                |                |                |                | 0.14<br>[2.45] | -0.15<br>[-1.85] |
| # months        | 715            | 715            | 720            | 715            | 720            | 630            | 623              |
| $\bar{R}^2(\%)$ | 0              | 1              | 0              | 1              | 0              | 1              | 0                |

**Table 5:** Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EPI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form:  $r_t^{EPI} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$ , where  $X_k$  indicates each of the six most-closely related anomalies and  $f_j$  indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies,  $X$ , are Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

|                 |                  |                  |                  |                  |                  |                  |                   |
|-----------------|------------------|------------------|------------------|------------------|------------------|------------------|-------------------|
| Intercept       | 0.18<br>[2.33]   | 0.23<br>[2.97]   | 0.18<br>[2.35]   | 0.12<br>[1.63]   | 0.19<br>[2.48]   | 0.23<br>[2.84]   | 0.19<br>[2.43]    |
| Anomaly 1       | 30.92<br>[8.10]  |                  |                  |                  |                  |                  | 13.48<br>[2.64]   |
| Anomaly 2       |                  | 19.34<br>[6.52]  |                  |                  |                  |                  | 12.96<br>[3.32]   |
| Anomaly 3       |                  |                  | 33.34<br>[8.18]  |                  |                  |                  | 38.69<br>[6.22]   |
| Anomaly 4       |                  |                  |                  | 27.98<br>[6.86]  |                  |                  | 14.80<br>[2.93]   |
| Anomaly 5       |                  |                  |                  |                  | 13.83<br>[3.57]  |                  | -18.62<br>[-3.26] |
| Anomaly 6       |                  |                  |                  |                  |                  | 7.35<br>[1.96]   | -17.37<br>[-3.79] |
| mkt             | -0.07<br>[-0.04] | 2.16<br>[1.15]   | 0.29<br>[0.16]   | 0.43<br>[0.24]   | -1.06<br>[-0.57] | -0.41<br>[-0.20] | 0.05<br>[0.02]    |
| smb             | -0.17<br>[-0.07] | 1.75<br>[0.65]   | -2.46<br>[-0.95] | 0.45<br>[0.17]   | -1.52<br>[-0.57] | 5.96<br>[1.84]   | -1.23<br>[-0.40]  |
| hml             | -1.24<br>[-0.36] | -7.95<br>[-2.13] | -4.43<br>[-1.26] | 1.87<br>[0.53]   | -2.15<br>[-0.59] | 0.03<br>[0.01]   | -3.81<br>[-1.03]  |
| rmw             | -9.97<br>[-2.45] | -4.49<br>[-1.13] | 8.35<br>[2.38]   | -3.24<br>[-0.84] | 8.04<br>[2.21]   | 9.27<br>[2.10]   | 2.28<br>[0.49]    |
| cma             | 19.82<br>[3.54]  | 22.74<br>[3.98]  | 6.73<br>[1.04]   | 25.41<br>[4.61]  | 25.12<br>[3.83]  | 22.65<br>[3.76]  | -6.08<br>[-0.88]  |
| umd             | 2.81<br>[1.57]   | 5.76<br>[3.17]   | 3.65<br>[2.05]   | 2.40<br>[1.32]   | 4.59<br>[2.49]   | 5.15<br>[2.70]   | 3.79<br>[2.09]    |
| # months        | 728              | 728              | 732              | 728              | 732              | 630              | 626               |
| $\bar{R}^2(\%)$ | 22               | 20               | 22               | 20               | 16               | 12               | 24                |



**Figure 8:** Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as EPI. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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